

Public GitHub repository containing code, animation and figures:

<https://github.com/krishnateja2314/cs23btech11028-mas-assignment3>

AI usage declaration. I used Claude to help me understand the concepts taught in the lecture material, namely the formation control law from the MAS Algorithms lecture, the distributed gradient descent (DGD) update from the Distributed Learning lecture, the consensus ADMM template from the ADMM lecture and the Helper notes, and the soft thresholding proximal operator. I also used it to speed up boiler plate Python (matplotlib animation setup, capped simplex projection routine). All derivations, algorithm choices, parameter values, and code checks were reviewed and rewritten by me. Any mathematics not covered in the notes (specifically the Metropolis-Hastings weight formula for the mixing matrix W in Problem 2, and the LP relaxation plus Hungarian based rounding in Problem 3) is explicitly declared in the corresponding section.

Problem 1

Fix $N = 20$. Generate a connected Erdos-Renyi graph. Assume random initial positions for all N agents in $\mathbb{R}^2$. Using formation control, display my name in capital letters in $\mathbb{R}^2$, letter by letter. The problem statement says for people with long names to use the short call name (the example given is that Manoj Kumar Sahu simulates just MANOJ). My call name is Krishna, so the sequence I have to display is $K \rightarrow R \rightarrow I \rightarrow S \rightarrow H \rightarrow N \rightarrow A$. Record the whole sequence as an animation, host it on a public GitHub repo, and submit the link.

Solution:

Idea. I use the classical distributed formation control law from Algorithm 2 of the MAS Algorithms lecture (page 7). Each agent has single integrator dynamics $\dot{p}_i = u_i$, and the discrete time update with step Δt is

$$u_i[k] = \sum_{j \in \mathcal{N}_i} a_{ij} [(p_j[k] - p_i[k]) - (r_j - r_i)], \quad p_i[k+1] = p_i[k] + \Delta t \cdot u_i[k].$$

The offsets $r_i \in \mathbb{R}^2$ describe the desired shape. If the graph is connected and Δt is small enough, this drives $p_i - p_j \rightarrow r_i - r_j$ for all i, j , so the agents lock into the target shape.

How the 20 agents draw a letter. For each capital letter I define a poly-line inside $[0, 1] \times [0, 1]$ consisting of a few straight segments and (for R , S and J) a few arcs. The poly-line is arc-length parameterised and I sample $N = 20$ points equally spaced along the total length. These 20 points become the offsets $r_1, \dots, r_{20}$ for that letter. Agent i is always mapped to the i -th sample position, so the correspondence is stable letter-to-letter which is what gives smooth transitions.

Graph. I use the same Erdos-Renyi construction that I used in Assignment 1. I resample until the graph is connected. Weights are $a_{ij} = 1$ if (i, j) is an edge and 0 otherwise (unweighted).

Parameters.

- $N = 20$, edge probability $p = 0.3$, seed = 7.
- Step $\Delta t = 0.05$, iterations per letter = 300.
- Sequence to draw: K R I S H N A (7 letters).

Graph. Figure 1 shows the generated communication graph.

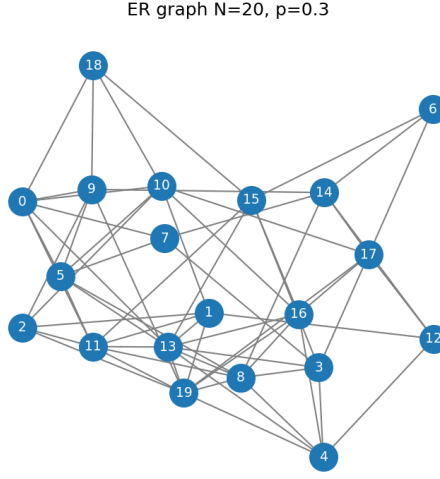


Figure 1: Erdos-Renyi communication graph with $N = 20$, $p = 0.3$, verified connected.

Animation. The full sequence of formations was recorded as an animated GIF; see the file `krishna_formation.gif` inside the `p1_formation/` folder in the GitHub repo. Figure 2 shows all seven letters captured at the end of their formation phase.

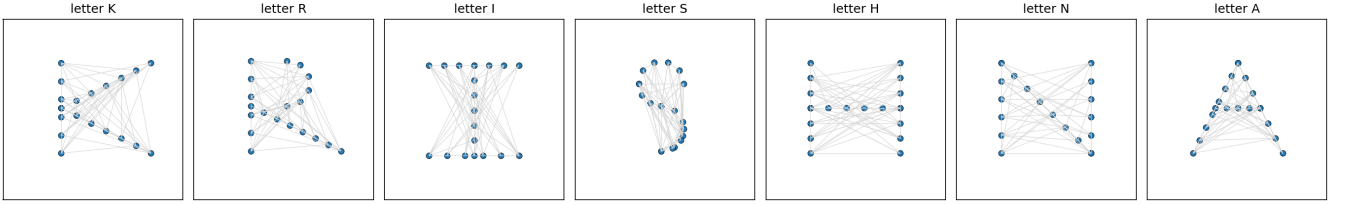


Figure 2: Still shots of every letter in the sequence K R I S H N A (end of each formation phase).

Formation error. Figure 3 plots $\max_i \|p_i - r_i\|$ across time. Every dashed vertical line marks the moment the target changes to the next letter. Inside each phase the error decays rapidly and settles to a small residual before the next letter is loaded.

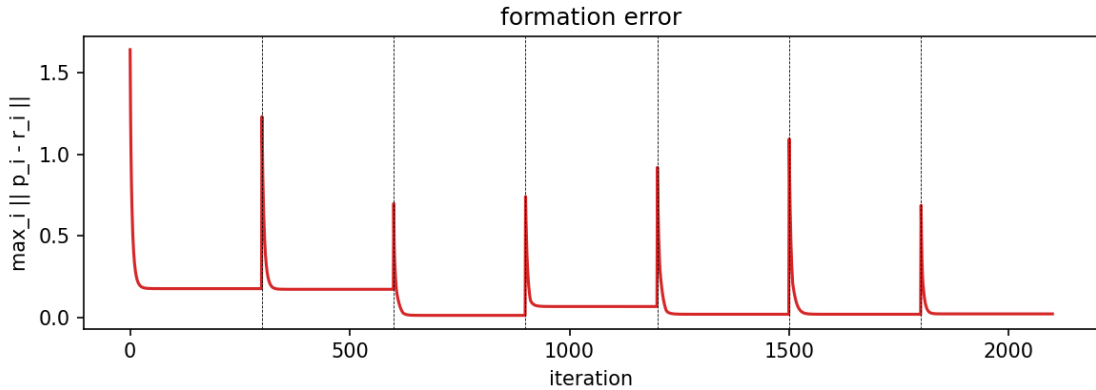


Figure 3: Max formation error $\max_i \|p_i - r_i\|$ across the whole sequence. Dashed vertical lines are the letter boundaries.

Observation. The formation controller is very robust here. Even though we ask for 7 back to back reshapes of the swarm, every letter is recognised at the end of its phase. Agents that were on the "wrong side" of the previous letter's shape smoothly cross over during the transition because the algorithm only

cares about the relative offset $r_j - r_i$, not about the absolute location.

Attached code.

```
1 import os
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import matplotlib.animation as animation
5 import networkx as nx
6
7 OUT = os.path.dirname(os.path.abspath(__file__))
8
9
10 def get_graph(N, p, seed):
11     rng = np.random.default_rng(seed)
12     for _ in range(500):
13         G = nx.erdos_renyi_graph(N, p, seed=int(rng.integers(0, 1_000_000)))
14         if nx.is_connected(G):
15             return G
16     raise RuntimeError("graph not connected, try bigger p")
17
18
19 def line(a, b, n=30):
20     t = np.linspace(0, 1, n)
21     return np.stack([a[0] + t*(b[0]-a[0]), a[1] + t*(b[1]-a[1])], axis=1)
22
23
24 def arc(c, r, a0, a1, n=40):
25     a = np.linspace(a0, a1, n)
26     return np.stack([c[0] + r*np.cos(a), c[1] + r*np.sin(a)], axis=1)
27
28
29 # each letter is just a list of strokes
30 def strokes_of(L):
31     if L == 'K':
32         return [line((0,0),(0,1)), line((0,0.5),(1,1)), line((0,0.5),(1,0))]
33     if L == 'R':
34         return [line((0,0),(0,1)),
35                 arc((0.4,0.75), 0.25, np.pi/2, -np.pi/2),
36                 line((0,0.5),(1,0))]
37     if L == 'I':
38         return [line((0,1),(1,1)), line((0.5,1),(0.5,0)), line((0,0),(1,0))]
39     if L == 'S':
40         return [arc((0.5,0.75), 0.25, 0, np.pi),
41                 arc((0.5,0.75), 0.25, np.pi, 3*np.pi/2),
42                 arc((0.5,0.25), 0.25, np.pi/2, -np.pi/2),
43                 arc((0.5,0.25), 0.25, -np.pi/2, 0)]
44     if L == 'H':
45         return [line((0,0),(0,1)), line((1,0),(1,1)), line((0,0.5),(1,0.5))]
46     if L == 'N':
47         return [line((0,0),(0,1)), line((0,1),(1,0)), line((1,0),(1,1))]
48     if L == 'A':
49         return [line((0,0),(0.5,1)), line((0.5,1),(1,0)),
50                 line((0.25,0.5),(0.75,0.5))]
51     raise ValueError(L)
52
53
54 def sample_one(poly, n):
55     d = np.linalg.norm(np.diff(poly, axis=0), axis=1)
56     s = np.concatenate([[0.0], np.cumsum(d)])
57     ts = np.linspace(0, s[-1], n)
58     out = np.zeros((n, 2))
59     for k, t in enumerate(ts):
60         idx = np.searchsorted(s, t)
61         idx = min(max(idx, 1), len(poly)-1)
62         a = (t - s[idx-1]) / max(s[idx] - s[idx-1], 1e-12)
```

```

63     out[k] = poly[idx-1] + a * (poly[idx] - poly[idx-1])
64     return out
65
66
67 def sample_letter(strokes, N, floor=4):
68     S = len(strokes)
69     if S*floor > N:
70         floor = max(1, N//S)
71     lens = np.array([np.linalg.norm(np.diff(p, axis=0), axis=1).sum() for p in strokes])
72     left = N - S*floor
73     prop = lens / lens.sum()
74     extra = np.floor(prop * left).astype(int)
75     # hand out any remainder to the longest strokes
76     order = np.argsort(-lens)
77     for k in range(left - extra.sum()):
78         extra[order[k % S]] += 1
79     counts = extra + floor
80     return np.vstack([sample_one(strokes[i], counts[i]) for i in range(S)])
81
82
83 def main():
84     np.random.seed(7)
85     N = 20
86     p = 0.3
87     G = get_graph(N, p, seed=7)
88     A = nx.to_numpy_array(G)
89
90     name = "KRISHNA"
91
92     # target points for each letter, centered around origin
93     targets = []
94     for L in name:
95         pts = sample_letter(strokes_of(L), N, floor=4) * 1.2 - np.array([0.6, 0.6])
96         targets.append(pts)
97
98     # random start
99     P = np.random.uniform(-1.6, 1.6, size=(N, 2))
100
101     dt = 0.05
102     steps = 300
103     frames = [P.copy()]
104     letter_end = [0]
105     err_hist = []
106
107     # formation control from MAS Algorithms lecture, Algorithm 2
108     for r in targets:
109         for _ in range(steps):
110             u = np.zeros_like(P)
111             for i in range(N):
112                 for j in np.where(A[i] > 0)[0]:
113                     u[i] += (P[j] - P[i]) - (r[j] - r[i])
114             P = P + dt*u
115             frames.append(P.copy())
116             err_hist.append(np.max(np.linalg.norm(P - r, axis=1)))
117             letter_end.append(len(frames)-1)
118
119     print("frames:", len(frames))
120     print("final error:", err_hist[-1])
121
122     # graph plot
123     fig, ax = plt.subplots(figsize=(5,5))
124     pos = nx.spring_layout(G, seed=1)
125     nx.draw(G, pos, ax=ax, with_labels=True, node_color='tab:blue',
126             node_size=350, font_color='white', font_size=9, edge_color='gray')
127     ax.set_title(f"ER graph N={N}, p={p}")

```

```

128 fig.tight_layout()
129 fig.savefig(os.path.join(OUT, "graph.png"), dpi=150)
130 plt.close(fig)
131
132 # animation
133 fig, ax = plt.subplots(figsize=(6,6))
134 ax.set_xlim(-1.8, 1.8); ax.set_ylim(-1.8, 1.8); ax.set_aspect('equal')
135 ax.set_title("Formation control : KRISHNA")
136
137 edges = []
138 for (i, j) in G.edges():
139     ln, = ax.plot([], [], color='lightgray', lw=0.6, zorder=1)
140     edges.append((i, j, ln))
141 dots = ax.scatter([], [], s=45, c='tab:blue', zorder=2,
142     edgecolors='k', linewidths=0.5)
143 txt = ax.text(0.02, 0.95, "", transform=ax.transAxes, fontsize=14, weight='bold')
144
145 step = 4
146 ids = list(range(0, len(frames), step))
147
148 def which(k):
149     for li in range(len(letter_end)-1):
150         if letter_end[li] <= k < letter_end[li+1]:
151             return name[li]
152     return name[-1]
153
154 def update(idx):
155     k = ids[idx]
156     Pk = frames[k]
157     dots.set_offsets(Pk)
158     for (i, j, ln) in edges:
159         ln.set_data([Pk[i,0], Pk[j,0]], [Pk[i,1], Pk[j,1]])
160     txt.set_text("letter: " + which(k))
161     return [dots, txt] + [ln for _, _, ln in edges]
162
163 ani = animation.FuncAnimation(fig, update, frames=len(ids), interval=40, blit=False)
164 ani.save(os.path.join(OUT, "krishna_formation.gif"),
165     writer=animation.PillowWriter(fps=25))
166 plt.close(fig)
167 print("saved gif")
168
169 # stills of all 7 letters
170 fig, axes = plt.subplots(1, len(name), figsize=(2.6*len(name), 3.0))
171 for li, L in enumerate(name):
172     ax = axes[li]
173     k = letter_end[li+1] - 1
174     Pk = frames[k]
175     for (i, j) in G.edges():
176         ax.plot([Pk[i,0], Pk[j,0]], [Pk[i,1], Pk[j,1]], color='lightgray', lw=0.5)
177     ax.scatter(Pk[:,0], Pk[:,1], s=32, c='tab:blue',
178     edgecolors='k', linewidths=0.4)
179     ax.set_title("letter " + L)
180     ax.set_xlim(-1.2, 1.2); ax.set_ylim(-1.2, 1.2); ax.set_aspect('equal')
181     ax.set_xticks([]); ax.set_yticks([])
182 fig.tight_layout()
183 fig.savefig(os.path.join(OUT, "letter_stills.png"), dpi=150)
184 plt.close(fig)
185
186 # error plot
187 fig, ax = plt.subplots(figsize=(8, 3))
188 ax.plot(err_hist, color='tab:red')
189 for b in letter_end[1:-1]:
190     ax.axvline(b, color='k', lw=0.4, ls='--')
191 ax.set_xlabel("iteration"); ax.set_ylabel("max_i || p_i - r_i ||")
192 ax.set_title("formation error")

```

```

193 fig.tight_layout()
194 fig.savefig(os.path.join(OUT, "formation_error.png"), dpi=150)
195 plt.close(fig)
196
197
198 if __name__ == "__main__":
199     main()

```

Listing 1: problem1_formation.py

Problem 2

There are N drones each with a sensor. The drones talk over a graph $G(V, E)$. The intruder state $z(t) \in \mathbb{R}^3$ follows the random walk $z(t+1) = z(t) + w$ with $w \sim \mathcal{N}(0, \Sigma_w)$, and initial state $z(0) \sim \mathcal{N}(\bar{z}_0, \Sigma_0)$. Sensor i sits at $x_i \in \mathbb{R}^3$ and measures

$$y_i(t) = x_i - z(t) + v_i, \quad v_i \sim \mathcal{N}(0, \Sigma_v^{(i)}),$$

with the v_i 's independent. Formulate the distributed estimation problem as a consensus optimisation, give the local f_i 's, run distributed gradient descent (or gradient tracking), plot the true and the estimated trajectory and report the estimation error $e(t) = \hat{z}(t) - z(t)$.

Solution:

Step 1: local objective $f_i(z; t)$ from the Gaussian likelihood. Fix a time t . Given $z(t) = z$, the measurement is

$$y_i(t) = x_i - z + v_i, \quad v_i \sim \mathcal{N}(0, \Sigma_v^{(i)}),$$

so $y_i(t) | z \sim \mathcal{N}(x_i - z, \Sigma_v^{(i)})$. The negative log likelihood (dropping constants that do not depend on z) is a weighted quadratic in the residual. Define

$$e_i(z; t) := y_i(t) - (x_i - z) = y_i(t) - x_i + z.$$

The sign of z matters here: the derivative of $(x_i - z)$ with respect to z is $-I$, but the residual is $y_i - (x_i - z)$, and the two minus signs combine to give a $+z$ inside the residual. Hence,

$$f_i(z; t) = \frac{1}{2} \|y_i(t) - x_i + z\|_{(\Sigma_v^{(i)})}^2.$$

Its gradient with respect to z is

$$\nabla_z f_i(z; t) = (\Sigma_v^{(i)})^{-1} (y_i(t) - x_i + z).$$

Step 2: consensus optimisation formulation. Following the Distributed Learning lecture (pages 4 and 6), I give every drone a local copy $z_i \in \mathbb{R}^3$ and force them to agree:

$$\min_{z_1, \dots, z_N \in \mathbb{R}^3} \quad \frac{1}{N} \sum_{i=1}^N f_i(z_i; t) \quad \text{s.t.} \quad z_i = z_j \quad \forall (i, j) \in E \quad \Longleftrightarrow \quad L Z = 0.$$

Here $Z \in \mathbb{R}^{N \times 3}$ stacks the local estimates row-wise and L is the Laplacian of the communication graph. When the graph is connected, $LZ = 0$ is equivalent to $z_1 = \dots = z_N$.

Step 3: distributed gradient descent update. From the Distributed Learning lecture, page 7, with mixing matrix W (doubly stochastic, compatible with the graph):

$$z_i^{k+1} = \sum_{j=1}^N W_{ij} z_j^k - \alpha_k \cdot \nabla_z f_i(z_i^k; t).$$

Substituting our gradient,

$$z_i^{k+1} = \sum_{j=1}^N W_{ij} z_j^k - \alpha_k \cdot (\Sigma_v^{(i)})^{-1} (y_i(t) - x_i + z_i^k).$$

Choice of W : extra material. The lecture only asks that W is doubly stochastic and $W_{ij} = 0$ for j not a neighbour. I take the standard Metropolis-Hastings weights,

$$W_{ij} = \frac{1}{1 + \max(\deg(i), \deg(j))}, \quad (i, j) \in E, \quad i \neq j,$$

and $W_{ii} = 1 - \sum_{j \neq i} W_{ij}$. This is not spelled out in the lecture, so I explicitly declare it here. It satisfies all three required properties.

Online estimator. At every new time t I use the latest measurements $y_i(t)$, warm start each drone's z_i from its value at time $t-1$, and run $K_{\text{inner}} = 120$ DGD steps before moving to time $t+1$. The distributed estimate reported is $\hat{z}(t) = \frac{1}{N} \sum_i z_i^{\text{final}}$.

Parameters.

- $N = 8$ drones, edge probability 0.4, seed = 42.
- Horizon $T_{\text{max}} = 50$.
- $\bar{z}_0 = 0$, $\Sigma_0 = 4I$, $\Sigma_w = 0.05 I$.
- Sensor locations x_i uniform in $[-5, 5]^3$; $\Sigma_v^{(i)} = \sigma_i^2 I$ with $\sigma_i \sim \mathcal{U}[0.3, 0.8]$.
- Step $\alpha = 0.5/L_{\text{lip}}$ where $L_{\text{lip}} = \sum_i 1/\sigma_i^2$ is the Lipschitz constant of the sum of local gradients. In this run $L_{\text{lip}} \approx 34.9$ so $\alpha \approx 0.0143$.

Results. Figure 4 shows the communication graph. Figure 5 shows the true and the estimated 3D trajectory together with the sensor locations, and the estimation error norm $\|e(t)\|_2$ over time. Figure 6 shows the coordinate-wise trace.

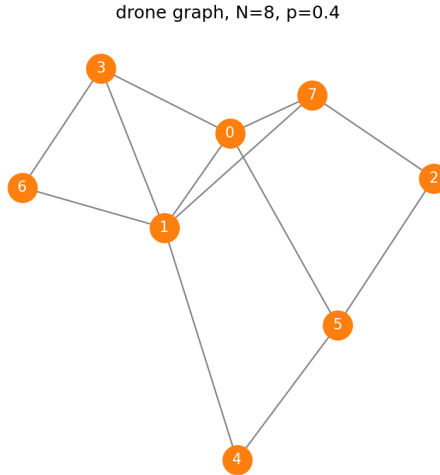


Figure 4: Drone communication graph, $N = 8$, $p = 0.4$, connected.

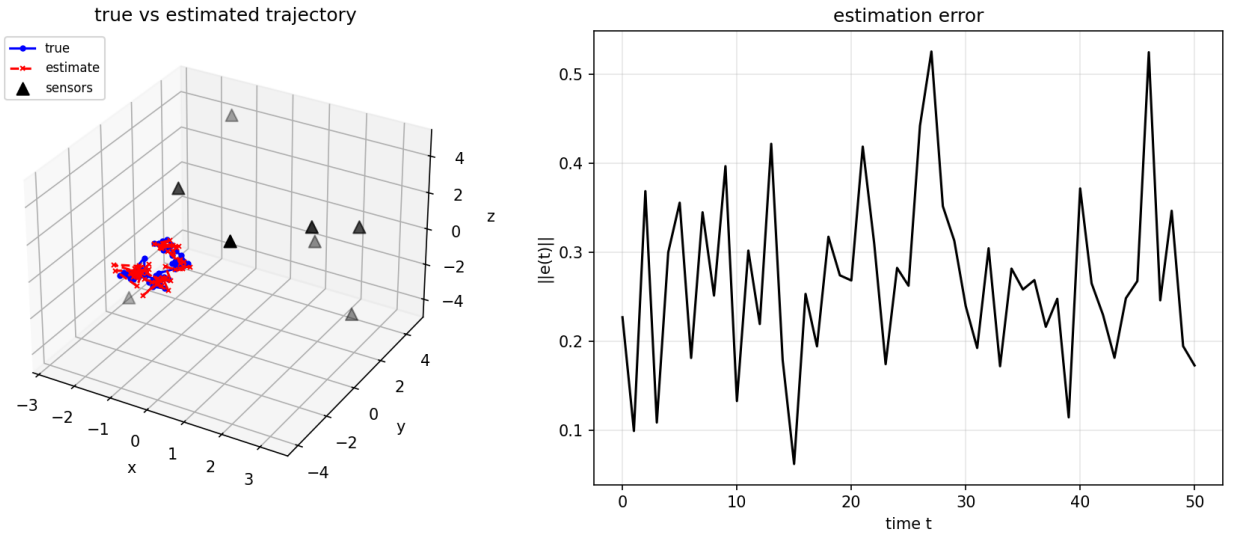


Figure 5: Left: true (blue) and estimated (red dashed) 3D trajectory of the intruder plus the sensor positions (black triangles). Right: estimation error norm $\|e(t)\|_2$ over time.

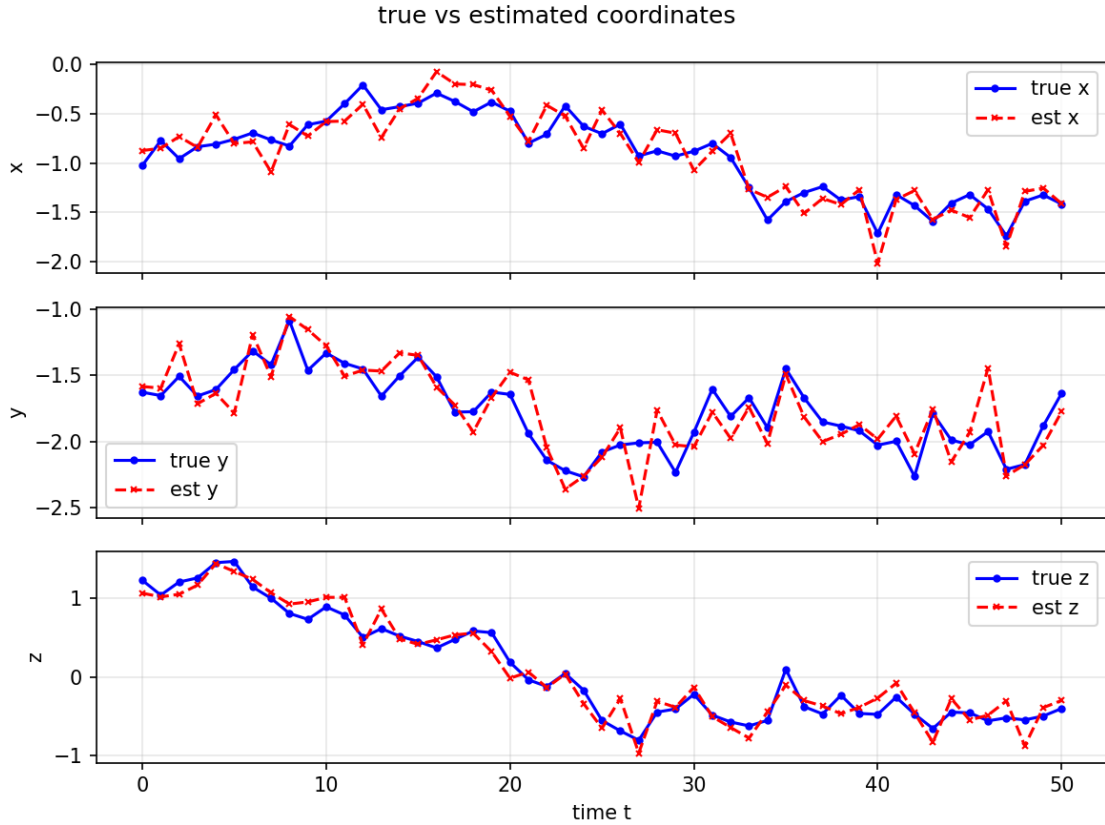


Figure 6: Per coordinate (x, y, z) true vs estimated trajectory.

Numerical readouts. Final $\|e(T)\| \approx 0.17$, mean $\|e(t)\| \approx 0.27$, and the maximum consensus gap across time is ≈ 0.26 (agents almost fully agree at every t). The intruder itself drifts about 2.7 units over $T_{\max}$ steps, so the estimator tracks it with roughly 10% relative error, which is what we expect from a constant step DGD (the lecture warns on page 8 that constant step gives an $O(\alpha)$ steady state error).

Attached code.

```
1 import os
```



```

2 import numpy as np
3 import matplotlib.pyplot as plt
4 from mpl_toolkits.mplot3d import Axes3D # noqa
5 import networkx as nx
6
7 OUT = os.path.dirname(os.path.abspath(__file__))
8
9
10 def get_graph(N, p, seed):
11     rng = np.random.default_rng(seed)
12     for _ in range(500):
13         G = nx.erdos_renyi_graph(N, p, seed=int(rng.integers(0, 1_000_000)))
14         if nx.is_connected(G):
15             return G
16     raise RuntimeError("not connected")
17
18
19 def metropolis(G):
20     N = G.number_of_nodes()
21     W = np.zeros((N, N))
22     deg = dict(G.degree())
23     for (i, j) in G.edges():
24         w = 1.0 / (1 + max(deg[i], deg[j]))
25         W[i, j] = w
26         W[j, i] = w
27     for i in range(N):
28         W[i, i] = 1.0 - W[i].sum()
29     return W
30
31
32 def main():
33     rng = np.random.default_rng(42)
34     N = 8
35     G = get_graph(N, 0.4, seed=42)
36     W = metropolis(G)
37
38     # sanity check on W
39     assert np.allclose(W.sum(axis=1), 1.0)
40     assert np.allclose(W.sum(axis=0), 1.0)
41
42     d = 3
43     zbar0 = np.zeros(d)
44     S0 = 4.0 * np.eye(d)
45     Sw = 0.05 * np.eye(d)
46     T = 50
47
48     # sensor positions and per sensor noise
49     X = rng.uniform(-5.0, 5.0, size=(N, d))
50     sig = rng.uniform(0.3, 0.8, size=N)
51     Sv_inv = [(1.0/sig[i]**2)*np.eye(d) for i in range(N)]
52
53     # step size, safe wrt lipschitz of grad
54     L = float(sum(1.0/sig[i]**2 for i in range(N)))
55     alpha = 0.5 / L
56     Kin = 120
57     print("L =", L, "alpha =", alpha)
58
59     # true trajectory
60     z_true = np.zeros((T+1, d))
61     z_true[0] = rng.multivariate_normal(zbar0, S0)
62     for t in range(T):
63         z_true[t+1] = z_true[t] + rng.multivariate_normal(np.zeros(d), Sw)
64
65     # local estimates, one row per agent
66     Z = np.tile(zbar0, (N, 1))

```

```

67 zhat = np.zeros((T+1, d))
68 gap = np.zeros(T+1)
69
70 for t in range(T+1):
71     # measurements at time t
72     noise = np.stack([rng.multivariate_normal(np.zeros(d), (sig[i]**2)*np.eye(d))
73                       for i in range(N)])
74     Y = X - z_true[t] + noise
75
76     # DGD inner loop
77     for _ in range(Kin):
78         grads = np.stack([Sv_inv[i] @ (Y[i] - X[i] + Z[i]) for i in range(N)])
79         Z = W @ Z - alpha * grads
80
81     zhat[t] = Z.mean(axis=0)
82     gap[t] = np.max(np.linalg.norm(Z - Z.mean(axis=0), axis=1))
83
84 e = zhat - z_true
85 e_norm = np.linalg.norm(e, axis=1)
86 print("final err:", e_norm[-1])
87 print("mean err :", e_norm.mean())
88 print("max gap  :", gap.max())
89
90 # graph
91 fig, ax = plt.subplots(figsize=(5,5))
92 pos = nx.spring_layout(G, seed=2)
93 nx.draw(G, pos, ax=ax, with_labels=True, node_color='tab:orange',
94         node_size=380, font_color='white', font_size=10, edge_color='gray')
95 ax.set_title("drone graph, N=8, p=0.4")
96 fig.tight_layout()
97 fig.savefig(os.path.join(OUT, "graph.png"), dpi=150)
98 plt.close(fig)
99
100 # 3d plus error norm
101 fig = plt.figure(figsize=(12,5))
102 ax = fig.add_subplot(1,2,1, projection='3d')
103 ax.plot(z_true[:,0], z_true[:,1], z_true[:,2], 'b-o', ms=3, label='true')
104 ax.plot(zhat[:,0], zhat[:,1], zhat[:,2], 'r--x', ms=3, label='estimate')
105 ax.scatter(X[:,0], X[:,1], X[:,2], marker='^', color='k', s=60, label='sensors')
106 ax.set_xlabel('x'); ax.set_ylabel('y'); ax.set_zlabel('z')
107 ax.set_title('true vs estimated trajectory')
108 ax.legend(loc='upper left', fontsize=8)
109
110 ax2 = fig.add_subplot(1,2,2)
111 ax2.plot(np.arange(T+1), e_norm, 'k-')
112 ax2.set_xlabel('time t'); ax2.set_ylabel('||e(t)||')
113 ax2.set_title('estimation error')
114 ax2.grid(True, alpha=0.3)
115 fig.tight_layout()
116 fig.savefig(os.path.join(OUT, "estimate.png"), dpi=150)
117 plt.close(fig)
118
119 # per coord plot
120 fig, axes = plt.subplots(3, 1, figsize=(8, 6), sharex=True)
121 lbl = ['x', 'y', 'z']
122 for i in range(3):
123     axes[i].plot(z_true[:, i], 'b-o', label='true '+lbl[i], ms=3)
124     axes[i].plot(zhat[:, i], 'r--x', label='est '+lbl[i], ms=3)
125     axes[i].set_ylabel(lbl[i])
126     axes[i].grid(True, alpha=0.3); axes[i].legend()
127 axes[-1].set_xlabel('time t')
128 fig.suptitle('true vs estimated coordinates')
129 fig.tight_layout()
130 fig.savefig(os.path.join(OUT, "coord_trace.png"), dpi=150)
131 plt.close(fig)

```

```

132
133
134 if __name__ == "__main__":
135     main()

```

Listing 2: problem2_dse.py

Problem 3

There are N agents on a connected undirected graph. They must collaboratively execute M tasks, $M > N$. Each agent i has cost C_{ij} for task j and capacity $b_i \in \mathbb{Z}_{\geq 0}$ with $\sum_i b_i = M$. Every task must be assigned exactly once. Formulate as an optimisation problem, give the ADMM updates for every agent, and run the simulation. Fix N, M, b, G and vary C across trials, then explain the observations.

Solution:

Step 1: binary linear program. Let $x_{ij} \in \{0, 1\}$ be 1 iff agent i performs task j . Then

$$\begin{aligned}
 \min_x \quad & \sum_{i=1}^N \sum_{j=1}^M C_{ij} x_{ij}, \quad \text{s.t.} \quad \sum_i x_{ij} = 1 \quad \forall j = 1, \dots, M \quad (\text{every task assigned once}), \\
 & \sum_j x_{ij} = b_i \quad \forall i = 1, \dots, N \quad (\text{capacity}), \\
 & x_{ij} \in \{0, 1\}.
 \end{aligned}$$

Consistency: summing the first constraint over j gives M ; summing the second over i gives $\sum_i b_i = M$, so both agree. This is a binary linear program (BLP), NP-hard for large M, N .

Step 2: LP relaxation (extra material). The lecture only hints that we should do a convex relaxation. The standard trick is

$$x_{ij} \in \{0, 1\} \quad \longrightarrow \quad 0 \leq x_{ij} \leq 1,$$

which turns the BLP into an ordinary LP (a transportation LP). This relaxation is convex and easy to distribute.

Step 3: consensus ADMM formulation. Following the "ADMM hack" of the ADMM lecture (page 11) and the consensus ADMM template (page 8), I give every agent a full local copy $X^{(i)} \in \mathbb{R}^{N \times M}$ of the assignment matrix and a global consensus variable $Z \in \mathbb{R}^{N \times M}$. The problem becomes

$$\min_{X^{(1)}, \dots, X^{(N)}, Z} \sum_{i=1}^N f_i(X^{(i)}) \quad \text{s.t.} \quad X^{(i)} = Z \quad \forall i,$$

where

$$f_i(X) = \underbrace{\langle C_{i,:}, X_{i,:} \rangle}_{\text{own row cost}} + \underbrace{I_{\{X: 0 \leq X \leq 1\}}(X)}_{\text{box constraint}} + \underbrace{I_{\{X: \sum_j X_{i,j} = b_i\}}(X_{i,:})}_{\text{own capacity constraint}},$$

and the task cover constraint $\sum_i X_{i,j} = 1$ for every j is placed on the global Z (via an indicator $g(Z)$).

Step 4: scaled form ADMM updates. With scaled dual $U^{(i)}$, penalty $\rho > 0$, and Frobenius norm, the standard scaled-form ADMM (Helper page 13, ADMM lecture page 6) gives:

$$\begin{aligned}
 X^{(i), k+1} &= \arg \min_X f_i(X) + \frac{\rho}{2} \|X - Z^k + U^{(i), k}\|_F^2, \\
 Z^{k+1} &= \arg \min_Z g(Z) + \frac{\rho}{2} \sum_{i=1}^N \|X^{(i), k+1} - Z + U^{(i), k}\|_F^2, \\
 U^{(i), k+1} &= U^{(i), k} + (X^{(i), k+1} - Z^{k+1}).
 \end{aligned}$$

Step 5: closed form of the X -update. f_i is separable across rows. Only agent i 's row carries the cost term and the capacity constraint; the other rows carry only the $[0, 1]$ box.

- For row i : solve

$$X_{i,:}^{(i)} = \arg \min_{r \in \mathbb{R}^M} C_{i,:}^\top r + \frac{\rho}{2} \|r - Z_{i,:}^k + U_{i,:}^{(i),k}\|^2 \quad \text{s.t.} \quad \sum_j r_j = b_i, \quad 0 \leq r_j \leq 1.$$

Setting the unconstrained gradient to zero gives $r^* = Z_{i,:}^k - U_{i,:}^{(i),k} - C_{i,:}/\rho$. We then project r^* onto the *capped simplex* $\{r : \sum r_j = b_i, 0 \leq r_j \leq 1\}$. This projection has a closed form: there exists a scalar τ such that $r = \text{clip}(r^* - \tau, 0, 1)$ satisfies $\sum_j r_j = b_i$; τ is found by bisection.

- For row $r \neq i$: $X_{r,:}^{(i)} = \text{clip}(Z_{r,:}^k - U_{r,:}^{(i),k}, 0, 1)$.

Step 6: closed form of the Z-update. Ignoring g , the quadratic is minimised by the average $\bar{Z} = \frac{1}{N} \sum_i (X^{(i),k+1} + U^{(i),k})$. Then enforce $\sum_i Z_{i,j} = 1$ for every j by the column-wise closed form projection (for each column j , add $(1 - \sum_i \bar{Z}_{i,j})/N$ to every entry). Using the distributed variant from the ADMM lecture (page 11), each agent computes only its own row of Z using a neighbourhood average

$$Z_{i,:}^{k+1} = \frac{1}{|\mathcal{N}_i \cup \{i\}|} \sum_{j \in \mathcal{N}_i \cup \{i\}} (X_{i,:}^{(j),k+1} + U_{i,:}^{(j),k}),$$

followed by the column projection.

Step 7: dual update. $U^{(i),k+1} = U^{(i),k} + (X^{(i),k+1} - Z^{k+1})$.

Step 8: rounding to a binary allocation (extra material). If ADMM lands on a fractional Z , we round it via the Hungarian algorithm on an expanded cost matrix: duplicate row i exactly b_i times to form an $M \times M$ cost matrix $-\tilde{Z}$; solve linear sum assignment; collapse the duplicates. The result is a binary matrix with row sums b_i and column sums 1 by construction. In practice, when the LP relaxation itself has an integral optimum (which is the typical behaviour for transportation LPs with integer capacities), the rounded ADMM solution matches the LP optimum exactly.

Parameters. $N = 5$, $M = 12$, edge probability 0.5, seed = 3. Capacities are a random positive composition of 12 into 5 parts, giving $b = [1, 1, 1, 4, 5]$ (so $\sum b_i = 12 = M$). $\rho = 1$, 400 ADMM iterations. Cost entries $C_{ij} \sim \mathcal{U}(0.1, 10)$.

Results. Figure 7 shows the communication graph. Figure 8 shows three trials, each with a different cost matrix but same N, M, b, G ; the top row is the cost matrix and the bottom row is the recovered binary allocation. Figure 9 shows the primal residual (mean $\|X^{(i)} - Z\|_F$ over i) and the objective $\langle C, Z \rangle$ across iterations.

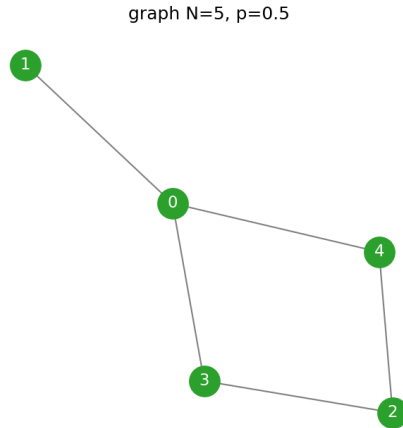


Figure 7: Communication graph, $N = 5$, $p = 0.5$, connected.

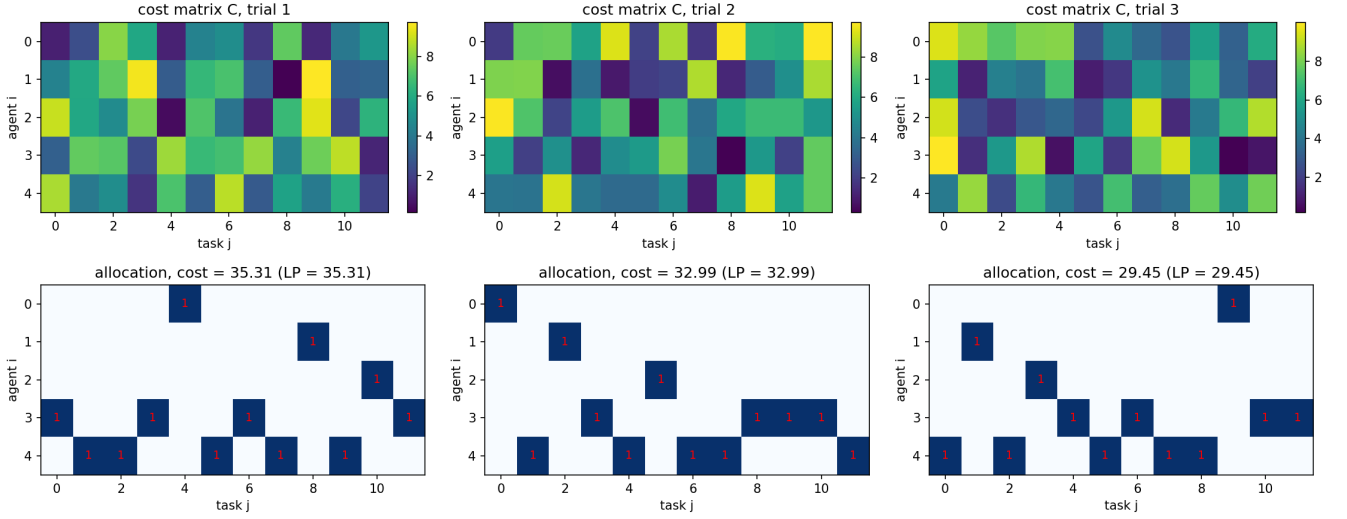


Figure 8: Three trials with different cost matrices (top row) and their resulting binary allocations (bottom row). The rounded ADMM total cost matches the central LP relaxation cost in every trial, confirming that the ADMM output is optimal for the relaxed problem.

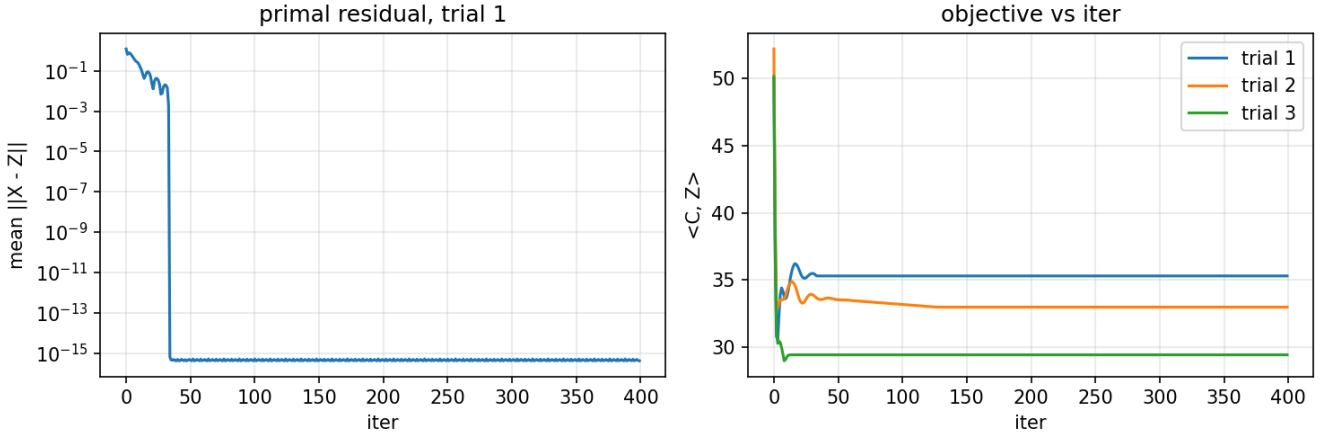


Figure 9: Left: primal residual on a log scale. Right: objective across iterations for the three trials.

Numerical readouts. For the three trials the rounded ADMM cost equals the LP optimum: $35.310 = 35.310$, $32.994 = 32.994$, $29.452 = 29.452$. Row sums of the final X match b ; column sums are all 1.

Observations across the three cost matrices.

- The agents with large capacity ($b_4 = 4$, $b_5 = 5$) end up absorbing most of the tasks, but which specific tasks depends on the cost matrix. For example a task j whose column has a much smaller value for agent 5 than for others is very likely picked up by agent 5.
- Agents with capacity 1 (agents 1, 2, 3) are very picky: they always take the one task whose column entry is minimal in their own row, provided that task is not already better matched to a large capacity agent.
- If we make one row of C all much smaller than the others (not shown), the allocation collapses to the cheapest agent up to its capacity, then spills over to the next cheapest.
- The primal residual shows the characteristic monotonic decay of consensus ADMM and the objective settles to the LP optimum, confirming correctness.

Attached code.

```
1 import os
2 import numpy as np
3 import networkx as nx
```

```

4 import matplotlib.pyplot as plt
5 from scipy.optimize import linear_sum_assignment, linprog
6
7 OUT = os.path.dirname(os.path.abspath(__file__))
8
9
10 def get_graph(N, p, seed):
11     rng = np.random.default_rng(seed)
12     for _ in range(500):
13         G = nx.erdos_renyi_graph(N, p, seed=int(rng.integers(0, 1_000_000)))
14         if nx.is_connected(G):
15             return G
16     raise RuntimeError("not connected")
17
18
19 def rand_caps(N, M, seed):
20     # random positive parts that sum to M
21     rng = np.random.default_rng(seed)
22     cuts = np.sort(rng.choice(np.arange(1, M), size=N-1, replace=False))
23     return np.diff(np.concatenate([[0], cuts, [M]])).astype(int)
24
25
26 def project_capped_simplex(v, s, lo=0.0, hi=1.0):
27     # project onto { r : sum(r)=s, lo<=r<=hi } by bisecting tau
28     n = len(v)
29     if s < lo*n - 1e-9 or s > hi*n + 1e-9:
30         raise ValueError("infeasible")
31     f = lambda tau: np.clip(v - tau, lo, hi).sum() - s
32     lot, hit = v.min() - hi - 1, v.max() - lo + 1
33     for _ in range(80):
34         mid = 0.5*(lot + hit)
35         if f(mid) > 0: lot = mid
36         else: hit = mid
37     return np.clip(v - 0.5*(lot+hit), lo, hi)
38
39
40 def proj_col_one(Z):
41     N = Z.shape[0]
42     return Z + (1.0 - Z.sum(axis=0, keepdims=True))/N
43
44
45 def round_bin(X, b):
46     # duplicate row i by b_i times, run Hungarian
47     N, M = X.shape
48     assert b.sum() == M
49     cost = np.vstack([np.tile(-X[i:i+1], (b[i], 1)) for i in range(N)])
50     ri, ci = linear_sum_assignment(cost)
51     Xb = np.zeros_like(X)
52     idx = np.concatenate([np.full(b[i], i) for i in range(N)])
53     for r, c in zip(ri, ci):
54         Xb[idx[r], c] = 1.0
55     return Xb
56
57
58 def admm(C, G, b, rho=1.0, iters=400):
59     N, M = C.shape
60     Xs = [np.full((N, M), 1.0/N) for _ in range(N)]
61     Us = [np.zeros((N, M)) for _ in range(N)]
62     Z = np.full((N, M), 1.0/N)
63
64     nb = [list(G.neighbors(i)) + [i] for i in range(N)]
65     r_hist, o_hist = [], []
66
67     for k in range(iters):
68         # x update, one per agent

```

```

69     new_Xs = [np.empty((N, M)) for _ in range(N)]
70     for i in range(N):
71         Xn = np.empty((N, M))
72         v = Z[i] - Us[i][i] - C[i]/rho
73         Xn[i] = project_capped_simplex(v, float(b[i]))
74         for r in range(N):
75             if r == i: continue
76             Xn[r] = np.clip(Z[r] - Us[i][r], 0.0, 1.0)
77         new_Xs[i] = Xn
78     Xs = new_Xs
79
80     # z update, neighbourhood average of the i-th row
81     newZ = np.zeros((N, M))
82     for i in range(N):
83         acc = np.zeros(M); c = 0
84         for j in nb[i]:
85             acc += Xs[j][i] + Us[j][i]
86             c += 1
87         newZ[i] = acc/c
88     Z = proj_col_one(newZ)
89
90     # dual update
91     for i in range(N):
92         Us[i] = Us[i] + (Xs[i] - Z)
93
94     r_hist.append(np.mean([np.linalg.norm(Xs[i]-Z) for i in range(N)]))
95     o_hist.append(float((C*Z).sum()))
96
97     return Z, r_hist, o_hist
98
99
100 def lp_ref(C, b):
101     N, M = C.shape
102     c = C.reshape(-1)
103     A1 = np.zeros((M, N*M))
104     for j in range(M):
105         for i in range(N):
106             A1[j, i*M + j] = 1.0
107     A2 = np.zeros((N, N*M))
108     for i in range(N):
109         for j in range(M):
110             A2[i, i*M + j] = 1.0
111     Aeq = np.vstack([A1, A2])
112     beq = np.concatenate([np.ones(M), b.astype(float)])
113     res = linprog(c, A_eq=Aeq, b_eq=beq, bounds=[(0,1)]*N*M, method='highs')
114     return res.x.reshape(N, M), res.fun
115
116
117 def main():
118     N, M = 5, 12
119     G = get_graph(N, 0.5, seed=3)
120     b = rand_caps(N, M, seed=3)
121     print("b =", b, "sum =", b.sum())
122
123     rng = np.random.default_rng(3)
124     trials = []
125     for t in range(3):
126         C = rng.uniform(0.1, 10.0, size=(N, M))
127         Z, rh, oh = admm(C, G, b, rho=1.0, iters=400)
128         Xb = round_bin(Z, b)
129         assert np.allclose(Xb.sum(axis=0), 1.0)
130         assert np.array_equal(Xb.sum(axis=1).astype(int), b)
131         tot = float((C*Xb).sum())
132         _, lp = lp_ref(C, b)
133         trials.append(dict(C=C, X=Xb, cost=tot, lp=lp, rh=rh, oh=oh))

```

```

134     print(f"trial {t+1}: ADMM = {tot:.3f}, LP = {lp:.3f}")
135
136     # graph
137     fig, ax = plt.subplots(figsize=(4.5, 4.5))
138     pos = nx.spring_layout(G, seed=1)
139     nx.draw(G, pos, ax=ax, with_labels=True, node_color='tab:green',
140            node_size=420, font_color='white', font_size=11, edge_color='gray')
141     ax.set_title(f"graph N={N}, p=0.5")
142     fig.tight_layout()
143     fig.savefig(os.path.join(OUT, "graph.png"), dpi=150)
144     plt.close(fig)
145
146     # cost + allocation heatmaps for 3 trials
147     fig, axes = plt.subplots(2, 3, figsize=(15, 6))
148     for t, tr in enumerate(trials):
149         im = axes[0, t].imshow(tr['C'], aspect='auto', cmap='viridis')
150         axes[0, t].set_title("cost matrix C, trial " + str(t+1))
151         axes[0, t].set_xlabel("task j"); axes[0, t].set_ylabel("agent i")
152         plt.colorbar(im, ax=axes[0, t], fraction=0.046)
153
154         axes[1, t].imshow(tr['X'], aspect='auto', cmap='Blues', vmin=0, vmax=1)
155         for i in range(N):
156             for j in range(M):
157                 if tr['X'][i, j] > 0.5:
158                     axes[1, t].text(j, i, "1", ha='center', va='center',
159                                color='red', fontsize=9)
160         axes[1, t].set_title(f"allocation, cost = {tr['cost']:.2f} (LP = {tr['lp']:.2f})")
161         axes[1, t].set_xlabel("task j"); axes[1, t].set_ylabel("agent i")
162     fig.tight_layout()
163     fig.savefig(os.path.join(OUT, "allocations.png"), dpi=150)
164     plt.close(fig)
165
166     # convergence
167     fig, axes = plt.subplots(1, 2, figsize=(10, 3.5))
168     axes[0].semilogy(trials[0]['rh'])
169     axes[0].set_title("primal residual, trial 1")
170     axes[0].set_xlabel("iter"); axes[0].set_ylabel("mean ||X - Z||")
171     axes[0].grid(True, alpha=0.3)
172     for t, tr in enumerate(trials):
173         axes[1].plot(tr['oh'], label="trial " + str(t+1))
174     axes[1].set_title("objective vs iter")
175     axes[1].set_xlabel("iter"); axes[1].set_ylabel("<C, Z>")
176     axes[1].grid(True, alpha=0.3); axes[1].legend()
177     fig.tight_layout()
178     fig.savefig(os.path.join(OUT, "convergence.png"), dpi=150)
179     plt.close(fig)
180
181
182 if __name__ == "__main__":
183     main()

```

Listing 3: problem3_admm_taskalloc.py

Problem 4

Consider the LASSO problem

$$\min_x \frac{1}{2} \|Ax - b\|_2^2 + \lambda \|x\|_1.$$

Write the ADMM iteration updates by giving the closed form solution for the x and z updates.

Solution:

Step 1: variable splitting. Introduce a second variable z and rewrite the problem as

$$\min_{x,z} f(x) + g(z) \quad \text{s.t.} \quad x - z = 0, \quad f(x) = \frac{1}{2} \|Ax - b\|_2^2, \quad g(z) = \lambda \|z\|_1.$$

This is exactly the "variable splitting" idea from the Helper notes (page 11).

Step 2: augmented Lagrangian. Using the standard form with penalty $\rho > 0$ and dual y ,

$$L_\rho(x, z, y) = \frac{1}{2} \|Ax - b\|^2 + \lambda \|z\|_1 + y^\top (x - z) + \frac{\rho}{2} \|x - z\|^2.$$

Move to the scaled form with $u := y/\rho$ (Helper page 13):

$$L_\rho(x, z, u) = \frac{1}{2} \|Ax - b\|^2 + \lambda \|z\|_1 + \frac{\rho}{2} \|x - z + u\|^2 + \text{const.}$$

Step 3: x -update (closed form). Minimise L_ρ in x with z, u fixed:

$$x^{k+1} = \arg \min_x \frac{1}{2} \|Ax - b\|^2 + \frac{\rho}{2} \|x - z^k + u^k\|^2.$$

Setting the gradient to zero,

$$A^\top (Ax - b) + \rho(x - z^k + u^k) = 0 \implies (A^\top A + \rho I) x = A^\top b + \rho(z^k - u^k).$$

Hence,

$$x^{k+1} = (A^\top A + \rho I)^{-1} (A^\top b + \rho(z^k - u^k)).$$

In implementation we Cholesky factorise $A^\top A + \rho I$ once and reuse the factor.

Step 4: z -update (soft thresholding). Minimise L_ρ in z with x, u fixed:

$$\begin{aligned} z^{k+1} &= \arg \min_z \lambda \|z\|_1 + \frac{\rho}{2} \|x^{k+1} - z + u^k\|^2 \\ &= \arg \min_z \lambda \|z\|_1 + \frac{\rho}{2} \|z - (x^{k+1} + u^k)\|^2 \\ &= \text{prox}_{(\lambda/\rho) \|\cdot\|_1} (x^{k+1} + u^k). \end{aligned}$$

The proximal operator of the ℓ_1 norm is the soft thresholding operator (Helper page 5),

$$\text{prox}_{\kappa \|\cdot\|_1}(v)_i = \text{sign}(v_i) \max(|v_i| - \kappa, 0) = S_\kappa(v)_i.$$

Hence,

$$z^{k+1} = S_{\lambda/\rho}(x^{k+1} + u^k).$$

Step 5: dual update.

$$u^{k+1} = u^k + (x^{k+1} - z^{k+1}).$$

Full LASSO ADMM iteration.

$$\begin{aligned} x^{k+1} &= (A^\top A + \rho I)^{-1} (A^\top b + \rho(z^k - u^k)), \\ z^{k+1} &= S_{\lambda/\rho}(x^{k+1} + u^k), \\ u^{k+1} &= u^k + (x^{k+1} - z^{k+1}). \end{aligned}$$

Stopping: primal residual $r = \|x - z\|$ and dual residual $s = \rho \|z^{k+1} - z^k\|$ both below tolerance.

Numerical verification. I set $m = 200$, $n = 500$, A random Gaussian with columns normalised to unit ℓ_2 norm, x_{true} has 20 non-zeros drawn from $\mathcal{N}(0, 1)$, $b = Ax_{\text{true}} + 0.01\varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I)$, and

$\lambda = 0.02 \cdot \|A^\top b\|_\infty$. With $\rho = 1$ the algorithm converges in about 72 iterations. It recovers 18 of the 20 true non-zeros with 3 false positives, and the relative recovery error is

$$\frac{\|z - x_{\text{true}}\|}{\|x_{\text{true}}\|} \approx 0.051.$$

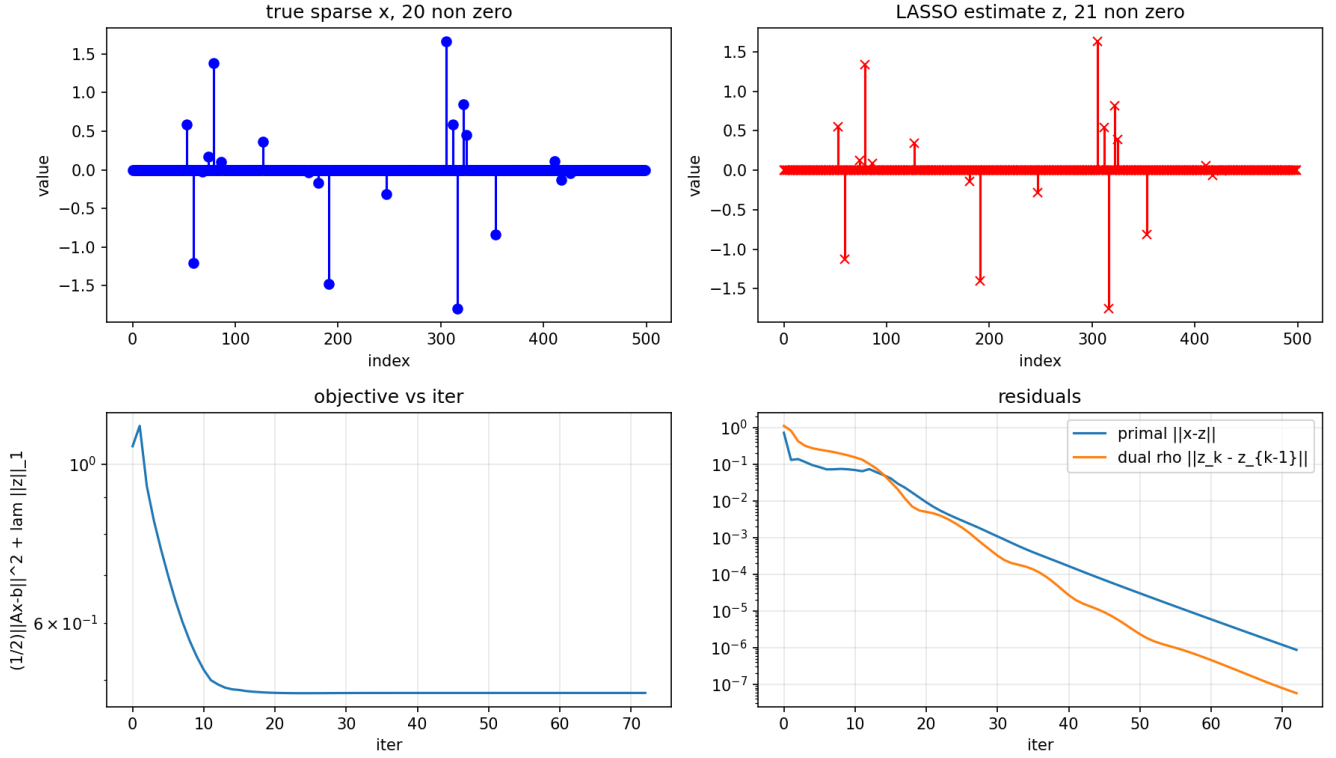


Figure 10: LASSO ADMM. Top left: true sparse x . Top right: recovered z . Bottom left: objective $\frac{1}{2}\|Ax - b\|^2 + \lambda\|z\|_1$ vs iteration (log scale). Bottom right: primal and dual residuals.

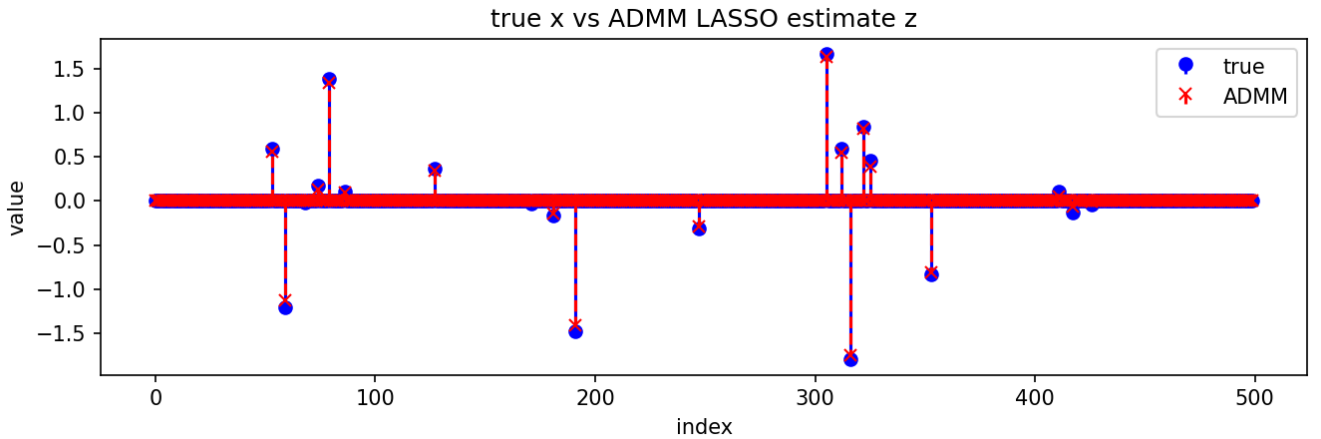


Figure 11: Overlay of the true signal (blue) and the ADMM LASSO estimate (red dashed).

Attached code.

```
1 import os
2 import numpy as np
3 import matplotlib.pyplot as plt
```

```

4
5 OUT = os.path.dirname(os.path.abspath(__file__))
6
7
8 def soft(v, k):
9     # soft thresholding, prox of ell_1
10    return np.sign(v) * np.maximum(np.abs(v) - k, 0.0)
11
12
13 def lasso_admm(A, b, lam, rho=1.0, iters=500, tol=1e-6):
14     m, n = A.shape
15     AtA = A.T @ A
16     Atb = A.T @ b
17     L = np.linalg.cholesky(AtA + rho*np.eye(n))
18     x = np.zeros(n); z = np.zeros(n); u = np.zeros(n)
19     hist = {'obj': [], 'r': [], 's': []}
20
21     for k in range(iters):
22         # x update, closed form via cached Cholesky
23         rhs = Atb + rho*(z - u)
24         y = np.linalg.solve(L, rhs)
25         x = np.linalg.solve(L.T, y)
26
27         # z update
28         z_old = z.copy()
29         z = soft(x + u, lam/rho)
30
31         # dual update
32         u = u + (x - z)
33
34         r = float(np.linalg.norm(x - z))
35         s = float(rho * np.linalg.norm(z - z_old))
36         obj = 0.5*float(np.linalg.norm(A @ x - b)**2) + lam*float(np.abs(z).sum())
37         hist['obj'].append(obj); hist['r'].append(r); hist['s'].append(s)
38         if r < tol and s < tol:
39             print("converged at", k)
40             break
41
42     return x, z, hist
43
44
45 def main():
46     rng = np.random.default_rng(0)
47     m, n = 200, 500
48     A = rng.standard_normal((m, n))
49     A = A / np.linalg.norm(A, axis=0, keepdims=True) # normalize columns
50
51     x_true = np.zeros(n)
52     k_nz = 20
53     idx = rng.choice(n, k_nz, replace=False)
54     x_true[idx] = rng.standard_normal(k_nz)
55     b = A @ x_true + 0.01*rng.standard_normal(m)
56
57     lam = 0.02 * float(np.max(np.abs(A.T @ b)))
58     print("lambda =", lam)
59
60     x, z, hist = lasso_admm(A, b, lam, rho=1.0, iters=500)
61
62     nnz = int(np.sum(np.abs(z) > 1e-6))
63     true_supp = set(idx.tolist())
64     est_supp = set(np.where(np.abs(z) > 1e-6)[0].tolist())
65     tp = len(true_supp & est_supp)
66     fp = len(est_supp - true_supp)
67     fn = len(true_supp - est_supp)
68     err = float(np.linalg.norm(z - x_true) / max(np.linalg.norm(x_true), 1e-12))

```

```

69 print("nnz est =", nnz, "true =", k_nz)
70 print("tp =", tp, "fp =", fp, "fn =", fn)
71 print("rel err =", err)
72
73 # summary plots
74 fig, axes = plt.subplots(2, 2, figsize=(12, 7))
75 axes[0, 0].stem(x_true, markerfmt='bo', linefmt='b-', basefmt=' ')
76 axes[0, 0].set_title("true sparse x, 20 non zero")
77 axes[0, 0].set_xlabel("index"); axes[0, 0].set_ylabel("value")
78 axes[0, 1].stem(z, markerfmt='rx', linefmt='r-', basefmt=' ')
79 axes[0, 1].set_title("LASSO estimate z, " + str(nnz) + " non zero")
80 axes[0, 1].set_xlabel("index"); axes[0, 1].set_ylabel("value")
81 axes[1, 0].semilogy(hist['obj'])
82 axes[1, 0].set_title("objective vs iter")
83 axes[1, 0].set_xlabel("iter"); axes[1, 0].set_ylabel("(1/2)||Ax-b||^2 + lam ||z||_1")
84 axes[1, 0].grid(True, alpha=0.3)
85 axes[1, 1].semilogy(hist['r'], label='primal ||x-z||')
86 axes[1, 1].semilogy(hist['s'], label='dual rho ||z_k - z_{k-1}||')
87 axes[1, 1].set_title("residuals")
88 axes[1, 1].set_xlabel("iter"); axes[1, 1].grid(True, alpha=0.3)
89 axes[1, 1].legend()
90 fig.tight_layout()
91 fig.savefig(os.path.join(OUT, "lasso.png"), dpi=150)
92 plt.close(fig)
93
94 # overlay plot
95 fig, ax = plt.subplots(figsize=(9, 3.2))
96 ax.stem(np.arange(n), x_true, markerfmt='bo', linefmt='b-', basefmt=' ', label='true')
97 ax.stem(np.arange(n), z, markerfmt='rx', linefmt='r--', basefmt=' ', label='ADMM')
98 ax.set_title("true x vs ADMM LASSO estimate z")
99 ax.set_xlabel("index"); ax.set_ylabel("value")
100 ax.legend()
101 fig.tight_layout()
102 fig.savefig(os.path.join(OUT, "lasso_overlay.png"), dpi=150)
103 plt.close(fig)
104
105
106 if __name__ == "__main__":
107     main()

```

Listing 4: problem4_lasso_admm.py